Raghuraman R

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EDUCATION

Chennai institute of Technology, Bachelor of Engineering, Mechatronics Engineering **2024**

CGPA- 8.69

Courses : Embedded systems, Microprocessor and Microcontrollers, Industrial Automation, Pneumatics and Hydraulics, Sensors and Instrumentation, Design of mechatronics systems

Navabharath Matriculation higher secondary school **2020**

HSC- 83%

Courses : Physics, Chemistry, Maths

PROFESSIONAL EXPIERENCE

Alstrut India Private limited, Chennai **2024 May – Present**

Controls and Automation Engineer

- Did own python library to program for palletizing application.
- Integrated ROS with Universal Robots and simulated using moveit2
- Handled Various types of End Of Line automation project
- Did PLCs and HMI programming for Siemens, Mitsubishi.
- Expertise in Universal Robot Advanced level of programming
- Prepared Electrical Bills of Material(BOM), Electrical Drawings, DFQ, DFM
- Created job estimation for the sales team

INTERNSHIPS

Alstrut India private limited, Chennai **2023 Dec - May**

Student Intern

- Hands on experience on Universal Robots
- Did support and programming in proface HMI, IPCs
- Provides support to various projects
- Worked on Ros2(humble) with universal robots driver
- Made a sample program of reading datas of a cobot using the socket communications through python

Projects support

- Uses universal robots script commands to scaling the laser distance sensor value and programmed the Cobot to move
- Take care of HMI Programming
- Take care of Cobot installation

TVS Sensing Solutions, Madurai 2022 Aug - Sep

Student intern

- Went through PLC programming ,Arduino and Injection moulding
- Build and programmed 6 DOF articulated robot as a prototype for pick and place application

SKILLS

Programming Languages: Python, C

PLC Programming Software Packages: Siemens TIA Portal, GxWorks3

HMI Programming Software Packages: GP Pro Ex

Design Packages: PC Schematics Automation

Communication Protocols: Profinet, TCP/IP, Modbus

Microcontrollers: Arduino, Esp8266, Esp32

Operating Systems: Windows, Ubuntu

Tools and Technologies: Ros2(Humble), Gazebo, Moveit2 ***(Basics)***

Robots Known: kuka kr-16-3, Melfa 4v-rfld, Universal Robot (both CB and E-series)

Other Devices: Proficient in operating and interfacing a following device including

- Laser Distance Sensor (Wenglor, Ifm)
- Tower lamp Interface Converter (Patlite)

PROJECTS & EXTRACURRICULAR

- **Smart Face shield for maintaining social distancing**

  Developed and implemented a Smart Face Shield prototype integrating an ultrasonic sensor to detect social distancing and an alert system using a buzzer. Addressed the challenges of social distancing during the COVID-19 pandemic by creating an innovative solution. Demonstrated problem-solving skills by integrating cutting edge technology into a wearable device, promoting safe practices, and raising awareness about social distancing guidelines. Showcased strong technical skills in sensor integration, circuit design, and Arduino programming. Utilized creativity and innovation to design a practical and functional solution that garnered positive results, effectively contributing to the community's safety and well-being.
- **Industrial fault monitoring system using IOT**

  Developed and implemented a "Smart Industry Fault Monitoring System" using ESP8266 and advanced sensors, revolutionizing industrial safety practices. Proactively detected faults in real-time through integration with IoT platforms, enabling remote monitoring and timely decision making. Successfully reduced downtime and optimized operations by leveraging predictive maintenance techniques, resulting in cost savings and improved operational efficiency.
- **Robotic arm**

  The project involved the construction and programming of a robotic arm using Arduino and servo motors. The robotic arm was designed to perform the task of picking up a wheel speed sensor from two conveyors and placing it in a designated box. Automated Sensor Placement: The primary objective of the project was to automate the process of picking up the wheel speed sensor from the conveyors and placing it in a box. The robotic arm was programmed to respond to signals from limit switches, which acted as manual controls for activating the arm's movement. Integration and Control: The project required integrating the Arduino board, servo motors, and limit switches to create a functional system. The Arduino board served as the control unit, executing the programmed instructions for the robotic arm's movement based on the input received from the limit switches. This allowed for precise control and coordination of the arm's actions during the sensor placement process.